

OmniMed-FL: A Robust Multimodal Federated Learning Framework for Clinical Diagnosis

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Abstract—We present OmniMed-FL, a controlled systems study of multimodal federated learning for five-class image-text classification. The proxy corpus contains 2,400 public chest radiographs, 600 synthetic COVID-19 image surrogates, and 3,000 class-conditioned synthetic notes paired to images by class rather than patient. Except for the labeled pooled-start ablation, runs initialize public DistilBERT and Vision Transformer Base/16 encoders with random task modules; no primary comparison starts from a pooled-data checkpoint. We vary Dirichlet skew, client count, anti-collapse components, initialization, eight operational fusion rules, missing text, and matched federated controls, while reporting GPU-synchronized training/aggregation time, peak allocated graphics processing unit (GPU) memory, and formula-derived 32-bit floating-point (FP32) bidirectional communication volume. At $K=5$ and severe skew ($\alpha=0.1$), local-only training obtains 0.310 macro-F1 score (five-client mean), versus 0.652 ± 0.057 for FedAvg, 0.685 ± 0.064 for FedProx, and 0.192 ± 0.129 for our SCAFFOLD-AdamW adaptation; the 0.033 FedProx-FedAvg gap is smaller than either two-seed sample standard deviation (SD). In the heterogeneity sweep, FedAvg changes from 0.896 ± 0.004 at $\alpha=5$ to 0.674 ± 0.185 at $\alpha=0.1$. A branch audit records 0.248/0.324/0.573 GiB one-way FP32 model-state sizes for text/image/multimodal models. Because all notes and the 600 COVID-19 images are synthetic, image-text pairing is not patient-level, and no external clinical validation is performed, these are descriptive proxy comparisons rather than estimates of diagnostic performance, privacy, or deployment readiness.

Index Terms—Multimodal Federated Learning, Non-Identically Distributed Data, Multimodal Fusion, Clinical Classification, Communication Cost, Retrieval

I. INTRODUCTION

CLINICAL prediction can combine radiographs and text, but cross-institutional centralized training ordinarily pools data. Federated learning (FL) instead exchanges parameter updates while clients retain raw examples [1]–[3]; this locality is not a privacy guarantee. Federated medical imaging [4]–[6] and centralized medical vision-language modeling are established, but their intersection raises sensitivity, collapse, fusion, and resource questions. OmniMed-FL studies these through a controlled five-class proxy benchmark. Because its public radiographs and synthetic notes are not patient matched, it tests the learning system, not clinical diagnosis. “Robust”

denotes stress tests across partitions, client counts, and missing modalities, not a certified clinical guarantee.

A. Contributions

We (i) evaluate heterogeneity, client count, anti-collapse components, initialization, and eight operational fusion rules; (ii) add matched FedMME-style one-shot and P-FIN-style missing-text controls; and (iii) report realized skew, GPU-synchronized training/aggregation time, peak memory, formula-derived bidirectional FP32 volume, and retrieval over all 600 validation queries.

II. RELATED WORK

Radiograph classifiers and Vision Transformers provide image-only precedents [1], [2], [7], [8]. Clinical FL studies cover FedAvg, multinational validation, and privacy-enhanced image learning [4]–[6]; FedProx and SCAFFOLD address heterogeneity [9], [10]. BiomedCLIP, LLaVA-Med, and MedCLIP are centralized medical image-text models [11]–[13]. Recent multimodal FL includes FedMME’s one-shot ensemble and P-FIN’s uncertainty-aware missing-feature imputation [14], [15]. Dataset, backbone, and budget differences preclude cross-paper score claims; our labeled controls are matched adaptations, not exact reproductions.

III. SYSTEM DESIGN

A. Architecture

Figure 1 shows the pipeline. WordPiece tokenizes notes to 128 tokens; images are resized to 224×224 and ImageNet-normalized. The DistilBERT [16] [CLS] vector is projected to $\mathbf{h}_t \in \mathbb{R}^{256}$ and mean-pooled ViT tokens to $\mathbf{h}_v \in \mathbb{R}^{512}$. Eight lightweight operators perform projected concatenation, residual cross-attention, gated residual fusion, dual-projection concatenation, one-layer decoder A, a distinct two-query/two-layer decoder B, non-residual 384-dimensional cross-attention, and a two-token Transformer encoder. Historical code keys resemble foundation-model names, but these are compact fusion rules rather than implementations of those models. Each feeds the same two-layer, five-class multilayer perceptron (MLP),

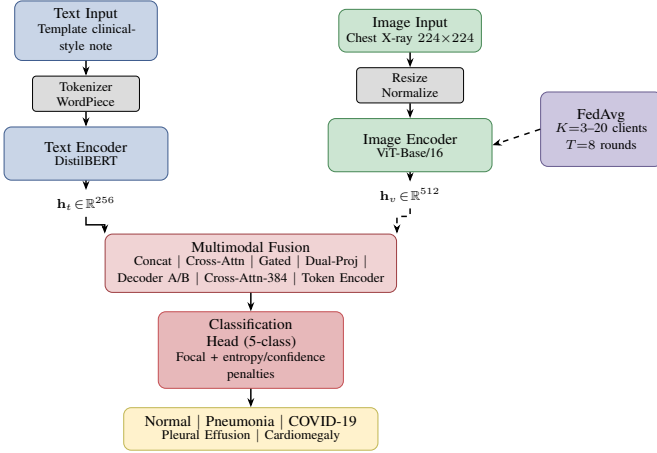


Fig. 1. OmniMed-FL architecture. DistilBERT and ViT-Base/16 encode the two proxy modalities; lightweight multimodal fusion combines them; FedAvg coordinates simulated clients without exchanging their raw examples.

and all eight use the same data, partition, optimizer, and local budget.

B. Federated and Local Objectives

Let K clients hold disjoint shards $\mathcal{D}_k$ of the training split, with $n_k = |\mathcal{D}_k|$ and $n = \sum_k n_k$, and let θ collect the trainable parameters on the fusion path: both encoders, the projection layers, the fusion operator and the classification head. The federated objective is the sample-weighted empirical risk

$$\min_{\theta} F(\theta) = \sum_{k=1}^K \frac{n_k}{n} F_k(\theta), \quad (1)$$

$$F_k(\theta) = \frac{1}{n_k} \sum_{(x,z,y) \in \mathcal{D}_k} \ell(f_{\theta}(x, z), y).$$

where x is a radiograph, z is its class-paired note, y is the class label, and f_{θ} is the fused five-class classifier. Shards are never exchanged; the server receives only the trainable tensors enumerated above.

For mini-batch $\mathcal{B}$, the implemented local loss is

$$\ell_{\mathcal{B}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} (1 - p_{i,y_i})^{\gamma} \text{CE}_{\epsilon,i} + \lambda \phi(h) + 5[q - 0.9]_+, \quad (2)$$

where p_{i,y_i} is the true-class probability, $\text{CE}_{\epsilon,i}$ is label-smoothed cross-entropy, $\gamma=2$, $\epsilon=0.05$, $C=5$, $h = H(\bar{p})/\log C$, $q = |\mathcal{B}|^{-1} \sum_i \max_c p_{ic}$, $\phi(h) = (1-h)[1 - \frac{8}{3}[h - 0.7]_+]_+$, and $[u]_+ = \max(u, 0)$. The entropy-diversity term grows under concentrated batch predictions but is attenuated above normalized entropy 0.7. Setting $\lambda=0$ removes only this term; the confidence penalty $5[q - 0.9]_+$ remains fixed. The class-balanced sampler and entropy-diversity term are the two anti-collapse components ablated in Section IV.

At each round, the server broadcasts $\theta^{(t)}$; each eligible client performs three local AdamW epochs using class-balanced mini-batches and (2), clips the gradient norm at one, and returns its trainable tensors. The server applies the sample-weighted update

in (1). Operational runs begin with public DistilBERT and ViT-Base/16 encoder weights plus random projection, fusion, and classification heads; no pooled-client model initializes them. Both encoders are fine-tuned, whereas unused unimodal heads and the inactive CNN fallback are excluded from optimization and communication.

C. Controlled Proxy Corpus

Image data. We form a balanced 3,000-image proxy corpus (600 examples per class). The public radiographs come from the Kermany pneumonia collection [17] and the NIH ChestX-ray dataset [18]. Loader logs verify 2,400 public images covering Normal, Pneumonia, Pleural Effusion and Cardiomegaly. The originally specified COVID-19 source was unavailable at execution time, so the 600 COVID-19 images are synthetic X-ray-style surrogates with bilateral opacity patterns. Images are therefore partly confounded by source collection and synthetic status. They are resized to 224×224 , normalized with ImageNet statistics, shuffled deterministically and split 80/20 into 2,400 training and 600 validation examples.

Clinical text data. No patient electronic health record (EHR) data are used. Each image label is paired at the class level with a synthetic clinical-style note generated from observation, symptom, context and indicator slots. Twenty-five percent of notes draw all slots from the target class, 35% mix target and non-target slots, and 40% are heavily mixed while retaining a probabilistic target-class indicator. This controlled construction reduces direct label wording, but the image and note are not from the same patient and must not be described as a real image-report pair.

What this corpus does and does not support. Every experimental arm sees the same controlled corpus, which supports internally controlled, descriptive comparisons between federated algorithms, fusion rules and ablations. The absolute scores are not diagnostic-performance estimates: class-conditioned templates can create shortcuts, label-level pairing removes real patient discordance and missingness, and the synthetic COVID-19 surrogates may not resemble clinical COVID-19 radiographs. The split is also not patient-grouped across source institutions. We therefore make no claim about clinical utility, deployment readiness, or generalization to real paired reports.

D. Experimental Setup

Hyperparameters. Each client uses AdamW with learning rate 10^{-4} and weight decay 0.01 for three local epochs per round. The reviewer suite uses eight rounds, batch size 16, FP32 tensors without automatic mixed precision (AMP), and an entropy-diversity weight of 1.0. The reference setting is five clients with Dirichlet $\alpha = 1.0$; Section IV varies α and the number of clients explicitly. Seeds 0 and 1 control pseudorandom initialization, sampling, and partitioning, except that pooled-start reuses one seed-0 checkpoint with seed-specific FL partitions. Experiments run in PyTorch on one NVIDIA H100 NVL device in a server shared with unrelated workloads. The GPU-synchronized timer covers sequential local training and aggregation within each round; validation, setup,

I/O, and networking are excluded, while server contention remains uncontrolled. Deterministic CUDA algorithms are not enforced, so seeds control software pseudorandom streams without guaranteeing bitwise-identical reruns. Simulated clients train sequentially on one GPU, so time is an execution-cost record rather than distributed throughput. Table I records the fixed protocol.

Partitioning. For each class, a Dirichlet draw allocates training indices across clients. Smaller α creates more concentrated client label distributions; $\alpha \rightarrow \infty$ approaches independent and identically distributed (IID) allocation [19]. Because nominal α does not fully describe a finite split, Fig. 3(f) shows per-client class shares for one severe-skew partition. The same split is fixed across rounds for a given seed. Shards with fewer than four examples skip local optimization but contribute the unchanged global state at their sample weight; weights are not renormalized over active clients. Scalability results therefore report active and nominal K .

Matched recent-method controls. Our matched FedMME-style arm uses 24 local epochs rather than FedMME’s native 100, uploads each client model once, and applies its equal-weight hard voting; our implementation breaks ties by mean softmax [14]. This preserves the common corpus, shard-size profile, encoders, optimizer, and matched local compute, but substitutes our five-class inputs for FedMME’s generated-report pipeline. The P-FIN-style stress test fixes three multimodal and two image-only clients, projects both encoders to 256-dimensional normalized features, and compares zero filling, deterministic FIN, Gaussian β -negative-log-likelihood (β -NLL) imputation ($\beta=0.5$), and uncertainty-weighted aggregation with Fed-UQ-Avg balance weight 0.6 and temperature 0.2 [15]. It uses the common concatenation classifier rather than P-FIN’s native fusion stack. FedProx uses $\mu=0.01$. The FedMME- and P-FIN-style controls are matched adaptations, not exact reproductions; their native scopes and communication schedules differ.

Rationale for Table I. Within each ablation or comparison family, only the named factor changes. Three local epochs and eight rounds give a tractable 24-local-epoch budget, while batch size 16 holds both encoders and one local model copy in GPU memory. The Cartesian systems grid crosses $K=\{3, 5, 10, 20\}$ with $\alpha=\{0.1, 1, 5\}$; the $K=5$ heterogeneity sweep additionally includes $\alpha=0.3$ and 0.5. Thus the study spans severe to near-IID label skew; $K=5$ is the small cross-silo reference and $K=3$ –20 probes count sensitivity. Batch size 16 is a conservative fixed choice for the largest FP32 model, not a throughput optimum. With only two seeds, the results support descriptive claims but not claims of statistical significance. The 10^{-4} rate, weight decay, precision, initialization, and $\lambda=1$ are fixed defaults, not per-model optima.

IV. RESULTS

A. Matched Performance and Ablations

At $\alpha=0.1$, local-only training averages a macro-F1 score of 0.310 across five clients in one partition. Across two matched

TABLE I
COMMON SIMULATION PARAMETERS.

Parameter	Value
Local Epochs (E)	3
Federated Rounds (T)	8
Batch Size	16
AdamW Learning Rate	1×10^{-4}
Weight Decay	0.01
Entropy-diversity weight (λ)	1.0
Systems-grid K (reference)	3, 5, 10, 20 (5)
Systems-grid α (reference)	0.1, 1, 5 (1)
$K=5$ additional α	0.3, 0.5
Seeds	0, 1
Initialization	Public encoders + random task modules
Precision	FP32 tensors (no mixed precision)

seeds, FedAvg obtains 0.652 ± 0.057 and FedProx obtains 0.685 ± 0.064 ; their 0.033 gap is smaller than both sample standard deviations (Fig. 2(a)). The SCAFFOLD–AdamW adaptation obtains 0.192 ± 0.129 ; its oscillation in panel (b) indicates instability in our AdamW control-variate adaptation, not classical SGD-based SCAFFOLD. The FedMME-style arm trains five client models independently for 24 epochs and combines predictions by equal-weight hard voting with mean-softmax tie-breaking; it preserves the one-shot idea but is not an exact reproduction.

Under severe skew, the full class-balanced-sampler/entropy-diversity stack reaches 0.716 ± 0.014 F1; removing the sampler, entropy-diversity term, or both gives 0.585 ± 0.148 , 0.701 ± 0.011 , and 0.657 ± 0.057 , respectively (panel (c)). The confidence penalty in (2) remains fixed in all four arms. Diversity diamonds distinguish transient loss of predicted-class diversity from lower F1; with only two seeds, no consistent directional component effect is evident at $\alpha=1$. Independent baseline, heterogeneity-sweep, and ablation runs nominally repeat the severe-skew FedAvg/concatenation setting but yield 0.652 ± 0.057 , 0.674 ± 0.185 , and 0.716 ± 0.014 ; these runs are reported separately rather than pooled or selected. Panel (d) trains all eight fusion rules in the operational FL loop.

The all-random arm retains the DistilBERT and ViT architectures but randomly initializes all encoder and task-module weights. All three initialization arms then use the same 8×3 FL schedule; pooled-start alone reuses one seed-0 checkpoint selected by a 12-epoch centralized pooled run for both FL seeds, so its extra pretraining is outside that budget. It starts at 0.843 rather than 0.704 round-1 F1 but ends at 0.818 rather than 0.839; only this labeled ablation uses the pooled checkpoint. Because that checkpoint was selected on the same validation split used for reporting, it is a validation-selected diagnostic, not an unbiased generalization estimate. The separate abort-off/on controls have epoch-12 F1 of 0.825 ± 0.001 and 0.843 ± 0.032 ; they are non-federated runs using pooled data without class-balanced sampling and with the entropy-diversity term disabled. Every run completes 12 epochs with minimum diversity 1.0, so the learning-rate boost/abort branch never fires; the difference is interpreted as rerun variation rather than an early-abort effect. Panel (f) is a matched P-FIN-style adaptation: 256-dimensional L_2 -normalized features and concatenation plus an MLP replace

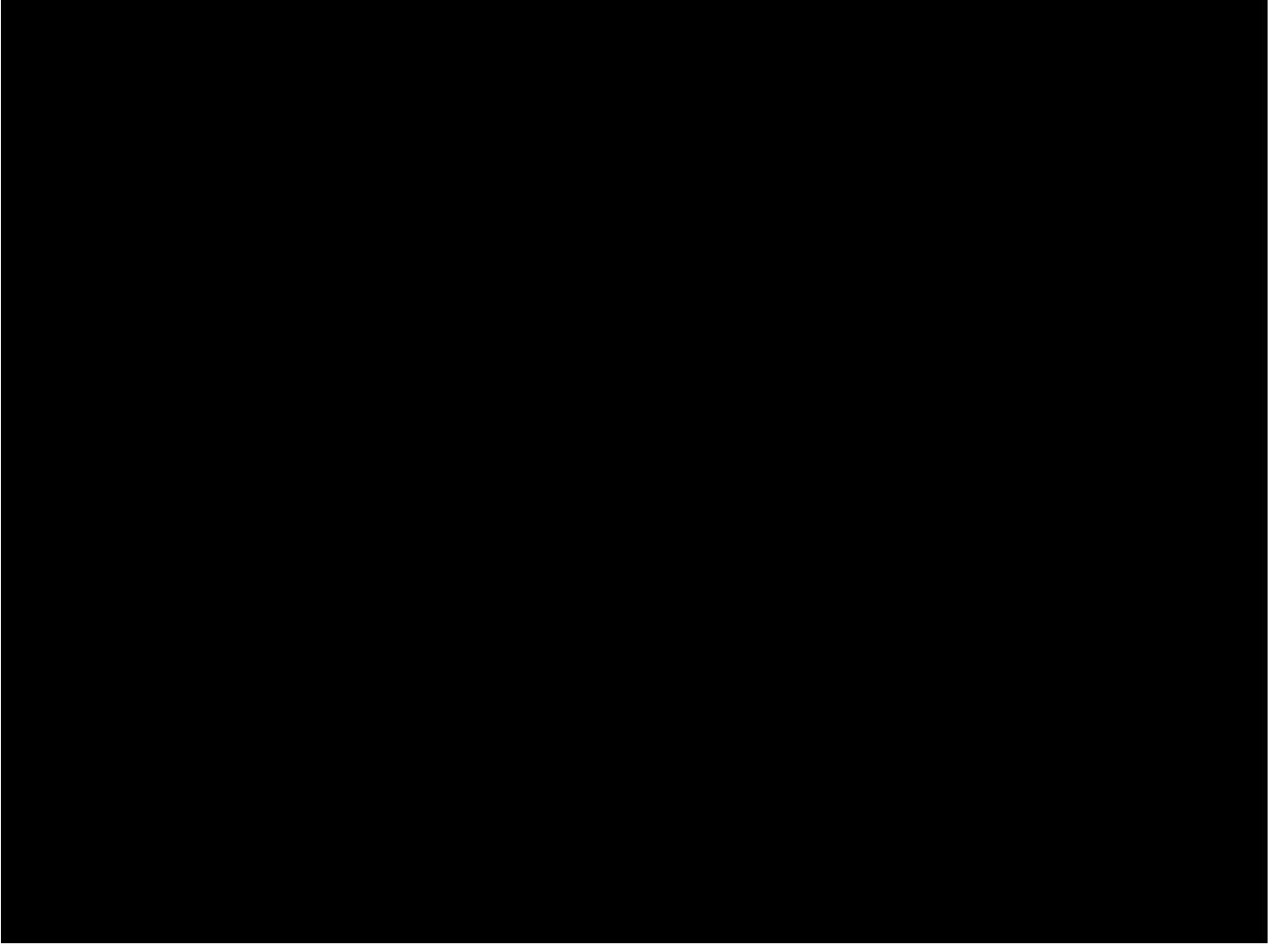


Fig. 2. Matched results (two-seed mean $\pm$ sample standard deviation unless noted). (a) Severe-skew baselines: local-only is the five-client mean from one partition; FedMME-style is a one-shot hard-voting ensemble after 24 independent local epochs, whereas iterative arms use 8 rounds of 3 epochs. (b) Iterative convergence. (c) Anti-collapse F1 with diamonds for minimum predicted-class diversity. (d) Eight operational fusion rules. (e) Federated initialization (round-1 diamonds) and separated, non-federated pooled abort controls whose bars show epoch-12 F1; their trigger never fires. (f) P-FIN-style adaptation with three multimodal and two image-only clients; all held-out text is forced missing for the primary metric, while full-text F1 is secondary.

the source bidirectional-attention stack; forced-missing text is primary and full-text performance secondary.

B. Scalability, Communication, and Resource Cost

Figure 3(a) assembles the Cartesian product of $K=\{3, 5, 10, 20\}$ and $\alpha=\{0.1, 1, 5\}$ from matched protocol runs: the validated $K=5$ heterogeneity sweep, the $\alpha=1$ client sweep, and newly executed cross-cells. These suites are not pooled. At $K=5$, F1 falls from 0.896 ± 0.004 at lower skew ($\alpha=5$) to 0.674 ± 0.185 under severe skew; auxiliary $\alpha=0.3$ and 0.5 checks yield 0.769 ± 0.043 and 0.824 ± 0.113 . This spread and the realized allocation in panel (f) show why nominal α and one seed are insufficient. Cells report A/K (or the two-seed A range); shards with fewer than four samples skip local optimization.

Panels (b)–(c) report the runtime and peak allocated memory of the GPU-synchronized sequential local-training and aggregation block on one shared H100 NVL; validation, setup,

input/output operations, networking, concurrency, and multi-node throughput are excluded. For $|\theta|$ trainable parameters, b bytes/parameter, T rounds, and nominal K , the plotted bidirectional volume is

$$V_{\text{nom}} = 2KT|\theta|b, \quad (3)$$

Here $|\theta|=153,935,621$, $b=4$, and $T=8$, so a 615,742,484-byte model state yields a nominal bidirectional volume of 27.5/45.9/91.8/183.5 GiB for $K=3/5/10/20$ (1 GiB = 2^{30} bytes). This nominal count remains when $A < K$; replacing K by A is an alternative estimate, not a measurement. Serialization, headers, retransmission, secure aggregation, compression, and algorithm-specific state are excluded.

In the one-run branch audit (panel (e)), text/image/multimodal F1 is 0.893/0.740/0.782; one-way FP32 model-state sizes are 0.248/0.324/0.573 GiB, sums of timed round durations are 18.5/33.6/45.0 min, and peak allocated memory is 1.69/3.18/4.77 GiB. The text branch outperforming

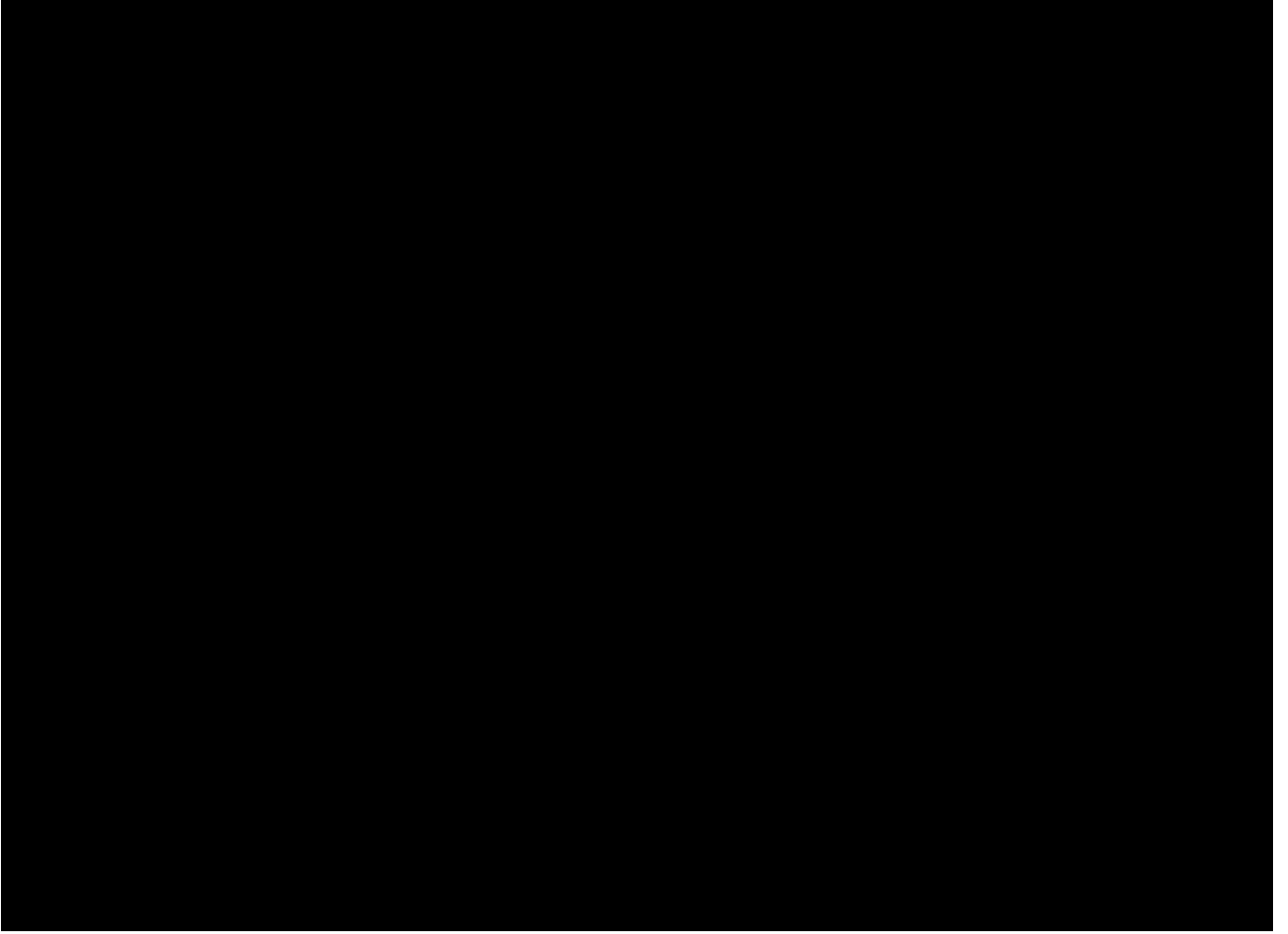


Fig. 3. Scalability/resource audit. Two-seed mean $\pm$ sample standard deviation is shown for (a) macro-F1 score (A/K is active/nominal clients or the two-seed A range), (b) the GPU-synchronized training/aggregation section per round, and (c) peak allocated memory within it. (d) Formula FP32 volume counts one download and upload for every nominal client in eight rounds, including locally skipped shards; it is not measured traffic. (e) One-run branch cost at $\alpha=1$, $K=5$ (row-normalized shading; absolute annotations show F1, timed-round sum, memory, and one-way model-state size). (f) Severe-skew seed-0 counts, colored by within-client class share.

multimodal concatenation shows that this proxy run does not support blanket multimodal superiority; class-conditioned template shortcuts are one plausible cause.

C. Retrieval-Stage Evaluation for Retrieval-Augmented Generation (RAG)

Exact inner-product FAISS search [20] over training-fitted TF-IDF note vectors yields condition-macro averages of 0.522 for top-1 same-label retrieval accuracy, 0.459 for same-label precision@5, and 0.707 for mean top-1 cosine similarity; the five condition rows cover all 600 validation queries (Fig. 4). This synthetic-note retrieval check is neither a fine-tuned DistilBERT result nor a generated-answer evaluation, and it does not establish clinical retrieval quality.

V. DISCUSSION AND LIMITATIONS

The experiments support systems-level conclusions only. Strong partition sensitivity persists despite anti-collapse measures: at $\alpha=0.1$, even identically configured FedAvg reruns vary

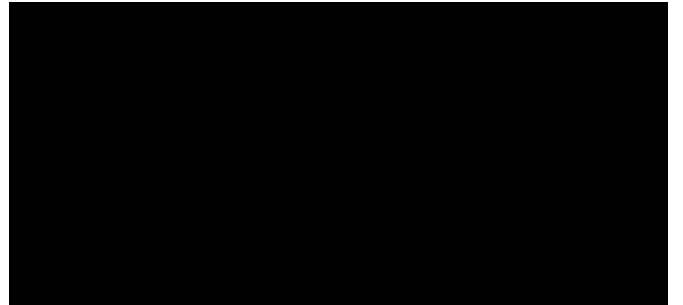


Fig. 4. FAISS exact inner-product retrieval over 2,400 indexed training-note vectors for 600 validation-note queries; TF-IDF is fitted on training notes only. Bars show top-1 same-label accuracy and same-label precision@5; diamonds show mean top-1 cosine similarity with population-standard-deviation whiskers. The dashed 0.2 line is a five-class label prior, not a similarity baseline; this evaluates retrieval, not generation.

substantially, and the audited runner skips local optimization for shards with fewer than four samples. The client-count

study is a sequential simulation; it does not model concurrent workers, network latency, stragglers, intermittent dropout, or heterogeneous accelerators. Robust aggregation against poisoned or Byzantine updates is also untested. Accordingly, “robust” in the title denotes the reported stress tests, not a formal guarantee.

Communication is a second boundary. Equation (3) exposes model-state scaling, while Fig. 3(e) shows the larger multimodal model state and memory footprint. A deployment study must additionally measure end-to-end traffic and latency under compression, partial participation, failures, and secure aggregation. This simulation implements neither secure aggregation nor the differentially private training and privacy accounting of Abadi *et al.* [21]. It also does not evaluate attacks on model updates; retaining raw examples locally is therefore not a formal privacy guarantee.

Clinical interpretation has additional limitations. All 3,000 notes and all 600 COVID-19 images are synthetic; the other 2,400 images are public radiographs. Image–text pairing is by class rather than patient; class-conditioned templates can encode label shortcuts; and label-level pairing does not reproduce the discordance, missingness, temporal context, or realistic cross-site variation of real records. Classes are partly confounded with source collection, and the split is grouped by neither patient nor source; acquisition artifacts may therefore occur on both sides of the split. There is no external or prospective validation. Thus neither absolute F1 nor retrieval scores estimate diagnostic safety, generalization, clinical utility, or deployment readiness.

Because only two seeds were run and deterministic CUDA algorithms were not enforced, the reported means and sample standard deviations are descriptive and do not support claims of statistical significance. The matched FedMME- and P-FIN-style arms retain selected method components under a common corpus and 24-local-epoch budget, but use different architectures and schedules and substitute our five-class inputs and fusion stack; they are not exact reproductions of the source papers. Larger multi-site datasets, more seeds, native-method replications, calibrated uncertainty, privacy/security tests, and prospective evaluation are necessary next steps.

VI. CONCLUSION

OmniMed-FL is a controlled systems study of multimodal FL rather than a clinical validation. Its reported performance and resource measurements are conditional on partition realization and protocol choices; the fusion, initialization, and missing-modality comparisons are descriptive stress tests. Real patient-paired multi-institutional studies and deployment safeguards remain necessary before clinical use.

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